Original Investigation

Effectiveness of Paliperidone Palmitate vs Haloperidol Decanoate for Maintenance Treatment of Schizophrenia
A Randomized Clinical Trial

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IMPORTANCE Long-acting injectable antipsychotics are used to reduce medication nonadherence and relapse in schizophrenia-spectrum disorders. The relative effectiveness of long-acting injectable versions of second-generation and older antipsychotics has not been assessed.

OBJECTIVE To compare the effectiveness of the second-generation long-acting injectable antipsychotic paliperidone palmitate with the older long-acting injectable antipsychotic haloperidol decanoate.

DESIGN, SETTING, AND PARTICIPANTS Multisite, double-blind, randomized clinical trial conducted from March 2011 to July 2013 at 22 US clinical research sites. Randomized patients (n = 311) were adults diagnosed with schizophrenia or schizoaffective disorder who were clinically assessed to be at risk of relapse and likely to benefit from a long-acting injectable antipsychotic.

INTERVENTIONS Intramuscular injections of haloperidol decanoate 25 to 200 mg or paliperidone palmitate 39 to 234 mg every month for as long as 24 months.

MAIN OUTCOME MEASURES Efficacy failure, defined as a psychiatric hospitalization, a need for crisis stabilization, a substantial increase in frequency of outpatient visits, a clinician's decision that oral antipsychotic could not be discontinued within 8 weeks after starting the long-acting injectable antipsychotics, or a clinician's decision to discontinue the assigned long-acting injectable due to inadequate therapeutic benefit. Key secondary outcomes were common adverse effects of antipsychotic medications.

RESULTS There was no statistically significant difference in the rate of efficacy failure for paliperidone palmitate compared with haloperidol decanoate (adjusted hazard ratio, 0.98; 95% CI, 0.65-1.47). The number of participants who experienced efficacy failure was 49 (33.8%) in the paliperidone palmitate group and 47 (32.4%) in the haloperidol decanoate group. On average, participants in the paliperidone palmitate group gained weight and those in the haloperidol decanoate group lost weight; after 6 months, the least-squares mean weight change for those taking paliperidone palmitate was increased by 2.17 kg (95% CI, 1.25-3.09) and was decreased for those taking haloperidol decanoate (−0.96 kg; 95% CI, −1.88 to −0.04). Patients taking paliperidone palmitate had significantly higher maximum mean levels of serum prolactin (men, 34.56 µg/L [95% CI, 29.75-39.37] vs 15.41 µg/L [95% CI, 10.73-20.08]; $P<.001$, and for women, 75.19 [95% CI, 63.03-87.36] vs 26.84 [95% CI, 13.29-40.40]; $P<.001$). Patients taking haloperidol decanoate had significantly larger increases in global ratings of akathisia (0.73 [95% CI, 0.59-0.87] vs 0.45 [95% CI, 0.31-0.59]; $P=.006$).

CONCLUSIONS AND RELEVANCE In adults with schizophrenia or schizoaffective disorder, use of paliperidone palmitate vs haloperidol decanoate did not result in a statistically significant difference in efficacy failure, but was associated with more weight gain and greater increases in serum prolactin, whereas haloperidol decanoate was associated with more akathisia. However, the CIs do not rule out the possibility of a clinically meaningful advantage with paliperidone palmitate.

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Long-acting injectable antipsychotic medications are prescribed to reduce nonadherence and relapse in people diagnosed with a schizophrenia-spectrum disorder. Long-acting injectable versions of older antipsychotic medications have been available for decades but their use has been limited in part due to their propensity to cause extrapyramidal symptoms, including tardive dyskinesia. Beginning in 1989, oral forms of newer antipsychotic medications, considered to entail lower risk of extrapyramidal symptoms, were introduced. Due to rapid acceptance of the newer oral antipsychotics, long-acting injectable versions of these medications were anticipated to gain widespread use. The first of these, risperidone microspheres, was introduced in 2003. Risperidone microspheres, however, must be refrigerated before use, reconstituted with a diluent provided by the manufacturer, and administered biweekly. In 2009, a long-acting version of risperidone's active metabolite, paliperidone palmitate, was brought to market. Paliperidone palmitate can be administered monthly and does not require refrigeration or reconstitution. Because of these logistical advantages, paliperidone palmitate was considered to be an important advance in long-acting injectable antipsychotic medications, although its high acquisition cost made its role uncertain.[1]

In recent years, head-to-head trials and meta-analyses have called into question the advantages of using atypical antipsychotic medications over older antipsychotics.[2-5] The CATIE schizophrenia trial (Clinical Antipsychotic Trials of Intervention Effectiveness) showed that when an older drug (perphenazine) was used at moderate doses, several newer ones were not superior in safety or effectiveness.[3] A recent secondary analysis provided evidence that perphenazine is not inferior to olanzapine, quetiapine, and risperidone with respect to symptom scores.[6] Moreover, some newer antipsychotic medications were shown to cause significant weight gain and to be associated with dyslipidemias and diabetes mellitus.[7,8]

This investigation compared the effects of long-acting injectable paliperidone palmitate and haloperidol decanoate, an older, widely used long-acting injectable antipsychotic. Based on an earlier comparison of oral risperidone to oral haloperidol decanoate,[9] we hypothesized that paliperidone palmitate would be associated with lower rates of efficacy failure and extrapyramidal symptoms than haloperidol decanoate, but that haloperidol decanoate would cause less weight gain and less increase in serum prolactin levels.

Method

Study Setting and Design

ACLAIMS (A Comparison of Long-acting Injectable Medications for Schizophrenia) was a multisite, parallel-group, double-blinded randomized clinical trial. The study was conducted at 22 US clinical sites affiliated with the National Institute of Mental Health–supported Schizophrenia Trials Network. Each site obtained institutional review board approval to conduct the study.

Patients

Patients were adults aged 18 to 65 years with a diagnosis of schizophrenia or schizoaffective disorder as defined by criteria from the *Diagnostic and Stastical Manual of Mental Disorders* (Fourth Edition, Text Revision; *DSM-IV-TR*) and confirmed by the Structured Clinical Interview for *DSM-IV*. Patients were eligible if judged by their clinician and study psychiatrist as likely to benefit from treatment with paliperidone palmitate or haloperidol decanoate and to be at risk of efficacy failure based on a history of medication noncompliance, significant substance abuse, or both. All patients demonstrated adequate decisional capacity to participate and provided written informed consent.

Patients with the following characteristics were excluded: currently stable and doing well using an antipsychotic regimen; not expected to benefit from the study medications due to past experience with risperidone, haloperidol, or paliperidone due to adverse effects or no improvement of severe symptoms in spite of an adequate treatment trial of at least 6 weeks' duration; moderate or severe tardive dyskinesia; presence of any medical condition that might preclude safe completion of the study; or intellectual disability. Women who were pregnant or breastfeeding were also excluded.

Patients attended a screening visit. If potentially eligible, a baseline visit was scheduled within 21 days. If determined eligible at the baseline visit, patients were then randomized on a 1:1 basis to paliperidone palmitate or haloperidol decanoate using an Internet-based system.

Interventions

A total of 353 patients enrolled for screening; 311 were found to be eligible and randomized to study treatment. Study treatments were long-acting injectable paliperidone palmitate supplied in dosages of 39 mg, 78 mg, 117 mg, 156 mg, and 234 mg; and injectable haloperidol decanoate supplied in vials of 50 mg/mL or 100 mg/mL. Each participant received a blinded trial of the oral version of the assigned medication prior to receiving an injection. In the case of paliperidone palmitate, the oral trial was with risperidone in accordance with the product label. The oral trial lasted from 4 to 7 days, with each patient recommended to receive 2 mg of either haloperidol or risperidone on days 1 and 2 and 4 mg thereafter. Haloperidol, 2 mg, and risperidone, 2 mg, were supplied in identical-appearing capsules. Oral benztropine, 1 mg, was supplied to treat extrapyramidal symptoms if needed. Patients who demonstrated allergy, extrapyramidal symptoms not relieved by benztropine, or other intolerability to the oral trial were dropped from the trial. Seventeen randomized patients never received the assigned long-acting injectable antipsychotic; only 2 of these were due to intolerability to the oral medication trial. The first injection was given 4 to 7 days after the baseline visit. Subsequent visits were at weeks 1, 2, 4, 6, 8, 10, and 12, then monthly (every 4 weeks) for up to 24 months.

Treatment condition was blinded from study physicians and all other personnel. Study physicians wrote orders for both of the potential long-acting injectable antipsychotic medications (eg, if haloperidol decanoate, administer 50 mg intramuscularly; if paliperidone palmitate, administer 117 mg intramuscularly). Each patient was then injected with only the randomly assigned drug. A clinician not otherwise

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involved in the trial administered the injection and concealed the identity of the medication from the patient and study personnel.

The loading strategy schedule described in the paliperidone palmitate prescribing information was recommended for both drugs. The recommended starting dose of paliperidone palmitate was 234 mg intramuscularly on day 1 followed on day 8 with 156 mg intramuscularly. The recommended standard monthly dose of paliperidone palmitate was 117 mg intramuscularly. The recommended starting dose of haloperidol decanoate was 50 mg intramuscularly on day 1 followed on day 8 with 50 mg intramuscularly. On day 28, the recommended dose of haloperidol decanoate was 75 mg intramuscularly, to be followed on day 56, and on subsequent monthly visits with 50 mg intramuscularly. The first 2 injections were given in the deltoid (day 1 and day 8). Subsequent monthly injections were given in the deltoid or gluteal muscle. The recommended injection schedules were adjusted according to the clinical situation. For the first 8 weeks, clinicians were allowed to supplement the long-acting injectable with any oral antipsychotic as needed.

If there were a desire to continue providing the study treatment drug to the patient when the following criteria were met: new-onset diabetes mellitus, weight gain of at least 15 pounds, increase in low-density lipoprotein (LDL) cholesterol of at least 20 mg/dL, worsening tardive dyskinesia, hospitalization, clinical worsening as indicated by the Clinical Global Impressions scale, or any serious adverse event, investigators were required to consult with the project's safety officer (a physician from whom treatment assignment was blinded). After the safety officer reviewed the case, study medication was continued if the clinician considered it in the best interest of the patient to continue and the patient and safety officer concurred.

Outcome Measures

The primary outcome was efficacy failure, which reflected inadequate control of the psychopathology of schizophrenia or schizoaffective disorder. Efficacy failure was determined for each study participant by an outcome adjudication committee consisting of 3 research psychiatrists who were blind to treatment assignment and not otherwise involved in the study. A majority vote of the committee determined whether and when a participant experienced efficacy failure. The criteria considered for efficacy failure included psychiatric hospitalization; a need for crisis stabilization; a clinically meaningful increase in frequency of outpatient visits; a clinician's decision that oral antipsychotic medication could not be discontinued within 8 weeks after starting the long-acting injectable; a clinician's decision to discontinue the assigned long-acting injectable due to inadequate therapeutic benefit; or, for patients successfully transitioned to receive a study long-acting injectable drug within 8 weeks, ongoing or repeated need for adjunctive oral antipsychotic medication.

Secondary outcome measures included change in weight from baseline and worst changes in fasting blood glucose, glycated hemoglobin, cholesterol, and triglycerides. The worst changes (eg, highest recorded level of triglycerides, lowest recorded levels of high-density lipoprotein [HDL]) were used for these laboratory-measured outcomes because interventions to treat abnormalities were allowed. For prolactin, the highest recorded level after baseline was the outcome. Other important secondary outcomes included measures of abnormal involuntary movements, akathisia, parkinsonism, and sexual functioning. Weight and measures of neurologic adverse effects were obtained at all study visits. Laboratory blood tests were obtained at screening, months 3 and 6, and then every 6 months. Patients were systematically queried about 12 adverse effects commonly associated with antipsychotic medications at each visit. Symptoms were measured using the Positive and Negative Syndrome Scale (PANSS) at baseline and then every 3 months. The scoring range for PANSS is 30 to 210, with higher scores reflecting greater severity of psychopathology.

Statistical Methods

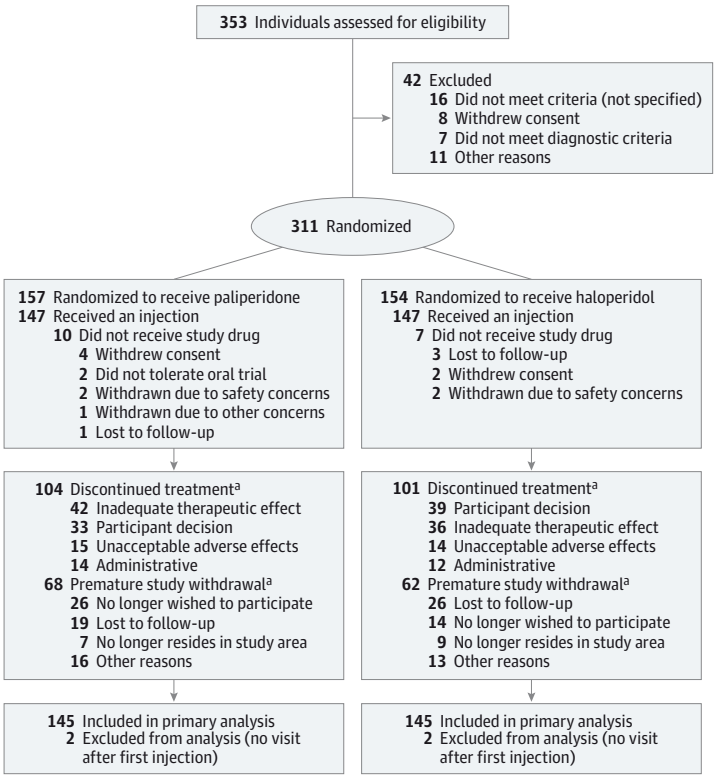
The primary analysis was conducted among the modified intent-to-treat population, which consisted of all patients who received at least 1 injection and at least 1 postbaseline assessment. The Kaplan-Meier method was used to estimate the proportion without efficacy failure by assessing time since first injection and using a site-stratified (2-sided) log-rank test for the primary comparison of time until efficacy failure. In the primary analysis, patients were censored 90 days after their last injection to account for the time most likely affected by the long-acting study medications. Planned supporting and sensitivity analyses included estimating the hazard ratio (HR) and 95% CI using a Cox proportional hazards model (controlling for baseline PANSS score and site); repeating the site stratified log-rank test without censoring 90 days after last injection; and conducting an unstratified log-rank test. The Cox model was expanded to test for site by treatment interaction and to test whether the HR for treatment was equal across 3 predefined time intervals: months 1 to 3, months 4 to 12, and months 13 to 24.[10] The site-stratified log-rank test was repeated for one subgroup defined a priori; participants who were not in an exacerbated state (ie, not hospitalized) at randomization. All main effects of treatment and treatment-by-site interactions for safety analyses were tested at the 2-sided $\alpha = .05$ level. For efficacy, interactions were tested at the $\alpha = .10$ level.

Safety analyses excluded data collected more than 6 weeks after a participant's last injection. Mixed-effect linear models (with spatial power covariance structure) were used to compare weight change over time. Fixed effects were included for assigned treatment, clinical site, baseline weight, time (months since first injection) and treatment-by-time interaction. The proportion of patients whose weight increased at least 15 pounds from baseline was compared using a Mantel-Haenszel χ^2 test. Analysis of covariance (ANCOVA) was used for metabolic analyses, with the worst-case change as the outcome and treatment, site, and baseline value as covariates.

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Figure 1. Participant Enrollment and Follow-up



[a] Subsets may not sum because some participants were counted among those who withdrew prematurely and also among those who discontinued treatment. Intent was to provide follow-up even after participants discontinued receiving injections, whether they withdrew from the study or not.

The same ANCOVA approach was used for comparisons of worst Abnormal Involuntary Movement Scale (AIMS) global score, Barnes Akathisia Scale (BAS) global score, and Simpson-Angus Extrapyramidal Scale (SAS) score. Incidence of clinically significant scores on 3 different assessments (ie, AIMS global score ≥2, BAS global score ≥3, and SAS score ≥1)[11] were compared between treatment groups using Mantel-Haenszel χ^2 tests, excluding patients who had a clinically significant score at baseline. As a posthoc analysis, the proportions meeting Schooler-Kane criteria for tardive dyskinesia (at least moderate dyskinetic movements in 1 body area or mild dyskinetic movements in 2 body areas)[12] were compared using Mantel-Haenszel χ^2 tests, excluding patients who met criteria at baseline.

For prolactin and associated adverse effects, separate analyses were planned for men and women. The key comparisons used analysis of variance (ANOVA) to compare the highest recorded prolactin level as the response, with treatment and site as covariates. Supporting analyses compared incidence of associated abnormalities (eg, gynecomastia or galactorrhea) between treatment groups using a Mantel-Haenszel χ^2 test or the Barnard exact test (if <10 events were observed) and highest Arizona Sexual Experiences (ASEX) scale score using ANOVA.

The original plan to randomize a total of 360 patients and follow-up for 2 years was modified due to resource constraints. The recruitment period was March 2011-July 2012; follow-up ended in July 2013. Ultimately, 311 individuals were randomized. The earliest enrollees were followed-up for as many as 24 months and the last enrollees for as many as 12 months. The planned sample size was expected to provide at least 80% power to detect a difference in survival curves (2-sided log-rank test, $\alpha = .05$), assuming efficacy failure rates of 0.56 and 0.40 for the haloperidol decanoate and paliperidone palmitate groups, respectively (ie, an HR of 1.6). Analyses were performed using SAS statistical software version 9.3 (SAS Institute Inc).

Results

Figure 1 summarizes the progress of patients who were screened and randomly assigned to each group. Baseline demographic and clinical characteristics of the 145 paliperidone palmitate and 145 haloperidol decanoate patients in the primary analysis are in **Table 1**. Patients were followed-up for a median of 488 days (25th-75th percentile, 225-645).

Dose

In the initial month of long-acting injectable treatment, which included doses on day 1 and day 8, the mean dose of paliperidone palmitate was 325 mg and of haloperidol decanoate 94 mg. Subsequently the mean monthly dose of paliperidone palmitate ranged from 129 to 169 mg and the mean monthly dose of haloperidol decanoate ranged from 67 to 83 mg.

Efficacy Failure

In the primary analysis, there was no statistically significant difference in the rate of efficacy failure for patients in the pali-

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Table 1. Baseline Demographic and Clinical Characteristics of Patients in the Modified Intent-to-Treat Population

Characteristics	Paliperidone Palmitate (n = 145)	Haloperidol Decanoate (n = 145)
Demographic		
Age, mean (SD), y	43 (12.6)	45 (12.3)
Sex, No. (%)		
Men	106 (73.1)	110 (75.9)
Women	39 (26.9)	35 (24.1)
Race, No. (%)[a]		
White	56 (38.6)	54 (37.2)
Black	83 (57.2)	83 (57.2)
Other[b]	6 (4.1)	8 (5.5)
Spanish, Hispanic, or Latino ethnicity, No. (%)[a]	6 (4.1)	8 (5.5)
Clinical		
In hospital, No. (%)	24 (16.6)	28 (19.3)
Weight, mean (SD), kg	90 (21.7)	90 (22.5)
BMI, mean (SD)[c]	30 (7.4)	30 (7.3)
HbA_{1c}, mean (SD),%	5.9 (1.3)	5.6 (0.6)
Blood glucose, mean (SD), mg/dL	104.0 (31.5)	94.6 (17.6)
Total cholesterol, mean (SD), mg/dL	179.7 (38.5)	181.5 (41.9)
LDL-cholesterol, mean (SD), mg/dL	104.6 (35.1)	108.1 (33.0)
Triglycerides, mean (SD), mg/dL	123.2 (85.4)	119.9 (80.5)
Prolactin in men, mean (SD), µg/L	17.4 (20.8)	17.8 (13.5)
Prolactin in women, mean (SD), µg/L	35.9 (35.2)	32.2 (38.5)
PANSS total score, mean (SD)[d]	73 (15.3)	70 (15.7)
CGI severity score, mean (SD)[d]	4.0 (0.8)	3.8 (0.9)
AIMS global severity score, median (range)[d]	0 (0-2)	0 (0-2)
SAS mean score, median (range)[d]	0 (0-1.5)	1.0 (0-1.5)
BAS global score, median (range)[d]	0 (0-3)	0 (0-3)
Psychiatric history, mean (SD)		
Age at first treatment for any behavioral or emotional problem, y	23 (9.3)	24 (10.9)
Age at first antipsychotic medication, y	26 (9.0)	27 (10.1)
SCID diagnoses, No. (%)		
Schizophrenia, lifetime	103 (71.0)	107 (73.8)
Schizoaffective disorder, lifetime	56 (38.6)	53 (36.6)
Major depression, past 5 y	36 (24.8)	40 (27.6)
Alcohol dependence, past 5 y	27 (18.6)	27 (18.6)
Alcohol abuse, past 5 y	42 (29.0)	43 (29.7)
Drug dependence, past 5 y	33 (22.8)	36 (24.8)
Drug abuse, past 5 y	50 (34.5)	45 (31.0)
Antisocial personality disorder, past 5 y	17 (11.8)	16 (11.0)

Abbreviations: AIMS, Abnormal Involuntary Movement Scale; BAS, Barnes Akathisia Scale; BMI, body mass index; CGI, Clinical Global Impressions Scale; HbA_{1c}, hemoglobin A_{1c}; LDL, low-density lipoprotein; PANSS, Positive and Negative Syndrome Scale; SAS, Simpson-Angus Scale; SCID, Structured Clinical Interview for *DSM-IV* (*Diagnostic and Statistical Manual of Mental Disorders* [Fourth Edition]).

SI conversions: To convert blood glucose to mmol/L, multiply by 0.0555; total and LDL cholesterol to mmol/L, multiply by 0.0259; triglycerides to mmol/L, multiply by 0.0113; and prolactin to pmol/L, multiply by 43.478.

[a] Race and ethnicity were self-reported.

[b] Includes American Indian or Alaska Native, Asian, Native Hawaiian or other Pacific Islander, and identification of 2 or more races.

[c] BMI is calculated as weight in kilograms divided by height in meters squared.

[d] Explanation of scores: PANSS (range, 30-210) higher scores reflect greater severity of psychopathology; CGI (range, 1 to 7) higher scores reflect greater severity of illness; AIMS global score (range, 0-4) higher scores reflect greater severity of abnormal movements; SAS mean score (range, 0-4) higher scores reflect greater severity of parkinsonisn symptoms; BAS global score (range, 0-3) higher scores reflect greater severity of akathisia.

peridone palmitate group (49 [33.8%]) vs those in the haloperidol decanoate group (47 [32.4%]; site-stratified log-rank $P = .90$; site and baseline PANSS adjusted HR, 0.98 [95% CI, 0.65-1.47]) (**Figure 2**). Results of all preplanned sensitivity and supporting analyses led to similar conclusions (eTable 1 in Supplement). Reasons for efficacy failure are in eTable 2 (in Supplement). The most common reasons for efficacy failure noted by the outcome ajudication committee were psychiatric hospitalization (44 [89.8%] of paliperidone palmitate events and 34 [72.3%] of haloperidol decanoate ones) and clinician discontinuation of study medication due to inadequate therapeutic effect (34 [69.4%] paliperidone palmitate events and 28 [59.6%] haloperidol decanoate ones).

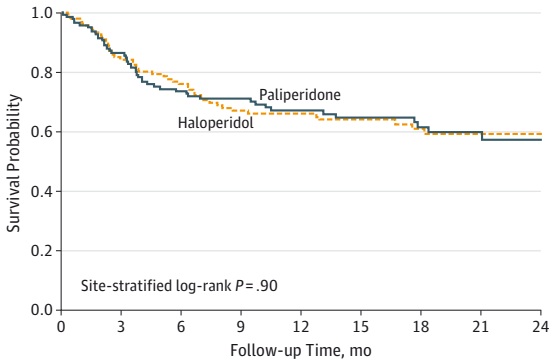
Secondary Outcomes

On average, participants taking paliperidone palmitate gained weight progressively over time, while those taking haloperidol decanoate lost weight. For example, at month 6, the least-squares mean weight change for participants in the paliperidone palmitate group was increased 2.17 kg (95% CI, 1.25 to 3.09) and for the haloperidol decanoate group was decreased −0.96 kg (−1.88 to −0.04) (**Table 2**). The test of time-by-treatment interaction showed statistically significant treatment group differences ($P < .0001$). Seven patients taking paliperidone palmitate (4.8%) compared with 2 (1.4%) in the haloperidol decanoate group discontinued treatment due to weight gain.

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Figure 2. Time to Efficacy Failure



No. at risk									
Paliperidone	145	119	91	76	62	51	36	26	10
Haloperidol	145	107	88	71	64	51	39	30	13
No. with event									
Paliperidone	0	19	36	39	43	45	47	48	49
Haloperidol	0	21	31	41	42	44	46	47	47

Adjusted hazard ratio for paliperidone palmitate vs haloperidol decanoate, 0.98 (95% CI, 0.65-1.47). Efficacy failure, determined by an outcome adjudication committee, reflected inadequate control of psychopathology.

There were no statistically significant differences between those treated with paliperidone palmitate and haloperidol decanoate in mean change to the highest recorded levels of glycated hemoglobin (HbA_{1c}), glucose, total cholesterol, LDL cholesterol, and triglycerides; or in the lowest recorded levels of HDL cholesterol.

There were no statistically significant differences in changes in ratings of abnormal involuntary movements as indicated by change from baseline score in the AIMS global score (0.43 [95% CI, 0.31-0.55] for paliperidone vs 0.50 [95% CI, 0.38-0.62] for haloperidol decanoate; P = .39; Table 2). There was no statistically significant difference in the incidence of probable tardive dyskinesia (15 in the paliperidone palmitate group [10.6%] and 21 in the haloperidol decanoate group [15.4%], P = .24). Participants taking haloperidol decanoate experienced greater increases in BAS global scores (0.45 [95% CI, 0.31-0.59] for those in the paliperidone palmitate group vs 0.73 [95% CI, 0.59-0.87] for the haloperidol decanoate group; P = .006). There was no statistically significant difference in changes in ratings of parkinsonism, as measured by the mean SAS score (0.21 [95% CI, 0.16-0.27] for the paliperidone palmitate group vs 0.25 [95% CI, 0.20-0.30] for the haloperidol decanoate group; P = .34). Fewer patients taking paliperidone palmitate than haloperidol decanoate started a medication to treat parkinsonism (18 [15.8%] vs 27 [29.3%]; P = .007) and akathisia (5 [3.6%] vs 16 [11.0%]; P = .03).

Treatment discontinuations due to neurologic adverse effects according to clinician judgment were as follows: 2 patients (1.4%) in the haloperidol decanoate group vs 1 (0.7%) in the paliperidone palmitate group due to akathisia; 3 (2.0%) in haloperidol decanoate group vs 1 (0.7%) in the paliperidone palmitate group due to parkinsonism; and 4 (2.7%) in the haloperidol decanoate group vs 1 (0.7%) in the paliperidone palmitate group due to tardive dyskinesia.

Among men, the highest mean prolactin level (SI unit conversion factor, multiply by 43.478 for pmol/L) was higher for paliperidone palmitate (34.56 µg/L; 95% CI, 29.75 to 39.37) than haloperidol decanoate (15.41 µg/L; 95% CI, 10.73 to 20.08) (P < .001) and in women, the highest mean prolactin level was higher for the paliperidone palmitate group (75.19 µg/L; 95% CI, 63.03 to 87.36) than haloperidol decanoate (26.84 µg/L; 95% CI, 13.29 to 40.40) (P < .001). There were no statistically significant differences in the proportions taking paliperidone palmitate or haloperidol decanoate who had a score on the Arizona Sexual Experiences scale of at least 19, which indicates sexual dysfunction for men or women. There were no significant differences in the incidence of gynecomastia or galactorrhea for men or women.

Overall, 68.0% of patients in the paliperidone palmitate group, compared with 59.9% of those in the haloperidol decanoate group, reported at least 1 adverse effect rated as moderate or severe (**Table 3**). Among the individual event types, 16.3% of patients taking paliperidone palmitate compared with 10.9% in the haloperidol decanoate group developed sialorrhea. This is the only adverse event with a difference of 5% or more between the groups. Seventy-six (51.7%) of patients in the paliperidone palmitate group experienced serious adverse events compared with 66 (44.9%) in the haloperidol decanoate group. One male participant in his sixties died of unknown causes approximately 6 weeks after his last haloperidol decanoate injection.

Decreases in PANSS total scores from baseline were similar for both groups at each time point (see eFigure 1 in Supplement). For example, at month 6, the least-squares mean PANSS change was −6.87 (95% CI, −8.79 to −4.94) for paliperidone palmitate and −6.40 (95% CI, −8.32 to −4.48) for haloperidol decanoate. In addition, as seen in Figure 1, rates of treatment discontinuation due to any cause (104/147 [70.7%] for the paliperidone palmitate group and 101/147 [68.7%] for the haloperidol decanoate group) and due to unacceptable adverse effects (15/147 [10.2%] for the paliperidone palmitate group and 14/147 [9.5%] for the haloperidol decanoate group) were similar.

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Table 2. Outcome Measures of Safety in the Modified Intent-to-Treat Population

Outcome	Paliperidone Palmitate (n = 147)	Haloperidol Decanoate (n = 147)	*P* Value[a]
Weight change (least-squares mean) from baseline, mean (95% CI), kg			
Month 6	2.17 (1.25 to 3.09)	−0.96 (−1.88 to −0.04)	<.001[b]
Month 12	3.46 (1.83 to 5.09)	−1.93 (−3.56 to −0.31)	
Month 18	4.75 (2.36 to 7.14)	−2.91 (−5.28 to −0.53)	
Month 24	6.04 (2.88 to 9.20)	−3.88 (−7.02 to −0.73)	
Ever gained ≥15 lbs from baseline, No. (%)	48 (33.1)	32 (22.4)	.03[c]
At least 1 laboratory assessment after first injection, No. of patients	**129**	**126**	
Laboratory values, worst change from baseline			
Results, least-squares mean (95% CI)			
HbA_{1c}, %	0.34 (0.17 to 0.52)	0.23 (0.06 to 0.41)	.38[d]
Blood glucose, mg/dL	21.13 (12.59 to 29.67)	20.96 (12.38 to 29.54)	.98[d]
Total cholesterol, mg/dL	12.42 (7.20 to 17.63)	16.82 (11.56 to 22.07)	.25[d]
LDL cholesterol, mg/dL	11.70 (7.06 to 16.34)	13.49 (8.85 to 18.14)	.59[d]
Triglycerides, mg/dL	36.91 (22.40 to 51.43)	46.57 (31.93 to 61.21)	.36[d]
HDL cholesterol, mg/dL	−5.28 (−6.74 to −3.83)	−4.52 (−5.98 to −3.05)	.47[d]
Neurologic effects			
AIMS global severity score			
Incidence of AIMS ≥2, No. (%)	28 (21.4)	30 (23.8)	0.57[c]
Worst change from baseline, least-squares mean (95% CI)	0.43 (0.31 to 0.55)	0.50 (0.38 to 0.62)	.39[d]
BAS global score			
Incidence of BAS ≥3, No. (%)	4 (2.8)	15 (10.6)	.006[c]
Worst change from baseline, least-squares mean (95% CI)	0.45 (0.31 to 0.59)	0.73 (0.59 to 0.87)	.006[d]
SAS mean score			
Incidence of SAS ≥1, No. (%)	109 (79.0)	101 (74.8)	.45[c]
Worst change from baseline, least-squares mean (95% CI)	0.21 (0.16 to 0.27)	0.25 (0.20 to 0.30)	.34[d]
Serum prolactin levels			
Among men only			
Highest level after baseline, least-squares mean (95% CI), µg/L	34.56 (29.75 to 39.37)	15.41 (10.73 to 20.08)	<.001[e]
Worst ASEX after baseline, least-squares mean (95% CI)[f]	17.68 (16.36 to 19.00)	17.95 (16.66 to 19.25)	.77[e]
ASEX score ≥19, No. (%)	34 (37.8)	37 (39.4)	.72[c]
Incidence of gynecomastia or galactorrhea, No. (%)[g]	5 (4.7)	3 (2.8)	.46[h]
Among women only			
Highest level after baseline, least-squares mean (95% CI), µg/L	75.19 (63.03 to 87.36)	26.84 (13.29 to 40.40)	<.001[e]
Worst ASEX after baseline, least-squares mean (95% CI)[f]	23.41 (21.01 to 25.80)	22.83 (20.12 to 25.54)	.75[e]
ASEX score ≥19, No. (%)	24 (72.7)	19 (73.1)	.88[c]
Incidence of gynecomastia, galactorrhea, or menstrual irregularities, No. (%)[i]	10 (38.5)	5 (29.4)	.13[c]

Abbreviations: AIMS, Abnormal Involuntary Movement Scale; ANCOVA, analysis of covariance; ANOVA, analysis of variance; ASEX, Arizona Sexual Side Effects; BAS, Barnes Akathisia Rating Scale; HDL, high-density lipoprotein; LDL, low-density lipoprotein; SAS, Simpson-Angus Scale.

[a] Comparison of paliperidone palmitate vs haloperidol decanoate.

[b] Test of time-by-treatment interaction.

[c] Comparison of binary outcomes is from a Cochran-Mantel-Haenszel test stratified by grouped site.

[d] Overall comparison between treatment groups obtained from ANCOVA, adjusting for baseline value and pooled site. The least-squares mean and standard error are from the corresponding ANCOVA model.

[e] Overall comparison between treatment groups obtained from ANOVA, adjusting for pooled site. The least-squares mean and standard error are from the corresponding ANOVA model.

[f] ASEX range is 6 to 30, with higher scores representing worse sexual functioning.

[g] Includes mild, moderate, and severe effects.

[h] From the Barndard exact test, due to low event counts in men.

[i] Incidence is among premenopausal women only and includes only moderate or severe effects (n=26 for the paliperidone group and n=17 for the haloperidol group).

Discussion

This randomized clinical trial found no evidence that long-acting injectable paliperidone palmitate was superior to haloperidol decanoate with respect to prevention of efficacy failure. However, based on the 95% CIs for the event rates, the results cannot rule out a clinically meaningful difference favoring one of the drugs.

Contrary to expectations, there was no statistically significant advantage for paliperidone palmitate when compared with haloperidol decanoate in ratings of the severity of abnormal involuntary movements and parkinsonism, or in the incidence of tardive dyskinesia. However, ratings of the se-

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Table 3. Adverse Events in the Intent-to-Treat Population

Outcome	No. (%)	
	Paliperidone Palmitate (n = 147)	Haloperidol Decanoate (n = 147)
Serious adverse events		
Any serious adverse event	53 (36.1)	45 (30.6)
Suicidal or homicidal ideation	23 (15.6)	21 (14.3)
Adverse events from systematic inquiry[a]		
Any moderate or severe adverse event	100 (68.0)	88 (59.9)
Insomnia	49 (33.3)	54 (36.7)
Sleepiness	41 (27.9)	44 (29.9)
Dry mouth	40 (27.2)	34 (23.1)
Increased appetite	33 (22.4)	26 (17.7)
Hypersomnia	24 (16.3)	20 (13.6)
Sialorrhea	24 (16.3)	16 (10.9)
Constipation	21 (14.3)	20 (13.6)
Orthostatic faintness	14 (9.5)	12 (8.2)
Incontinence/nocturia	13 (8.8)	8 (5.4)
Menstrual irregularities	12 (8.2)	5 (3.4)
Urinary hesitancy	7 (4.8)	9 (6.1)
Gynecomastia/galactorrhea	4 (2.7)	5 (3.4)

[a] Includes events rated moderate or severe.

verity of akathisia increased more for haloperidol decanoate, and more medications to manage akathisia and parkinsonism were started for patients in the haloperidol decanoate group, partially confirming that paliperidone palmitate has a lower propensity to cause extrapyramidal symptoms than haloperidol decanoate. The current study was informed by studies from the 1980s that compared standard doses of typical long-acting injectable antipsychotic medication with lower doses and found that patients' symptoms could be successfully controlled at these lower doses without relapse and without extrapyramidal toxicities.[13-15] Similarly, the CATIE schizophrenia trial found that modest doses of typical oral antipsychotic medication could be used effectively without excessive extrapyramidal symptoms.[3] The modest dose of haloperidol decanoate used here, approximately 75 mg intramuscularly per month, is lower than the equivalent oral dosage used in a trial that found an advantage of oral risperidone over oral haloperidol.[9] In that study, the mean (SD) daily dose of haloperidol decanoate was 11.7 (5.0) mg, whereas in this study, using a standard conversion from oral haloperidol to haloperidol decanoate of 10 to 15 times the daily dose, the corresponding daily dose is approximately 5.0 to 7.5 mg. The modest dosing of haloperidol decanoate in this study is consistent with current recommendations and may help to account for its better-than-expected comparative tolerability. It was unexpected that the relapse rate was similar to that in the earlier study, comparing oral risperidone with oral haloperidol, considering that the current study enrolled people at increased risk of nonadherence and relapse. One reason may be that the outcome in the oral trial, which was relapse, was somewhat broader than the definition of efficacy failure used here. Another possible reason is that long-acting injectable antipsychotic medications are more useful than oral ones in preventing relapse, but this question was not addressed in our study. The higher doses of haloperidol in the prior study may have had a negative effect on its tolerability and, consequentially, its effectiveness.

Early termination of the study's follow-up period, which meant that patients enrolled during the second year of the study were followed-up for at least 1 year but less than the planned 2 years, had little effect on statistical power for the primary outcome because the risk of efficacy failure during the second treatment year was low. However, the early termination may have resulted in less reliable estimates of weight change at later time points.

The study did not include a comparison with an oral antipsychotic medication. At the time this study was begun, 2 randomized clinical trials comparing oral and long-acting injectable antipsychotic medications were underway. Neither of these studies found an advantage of long-acting injectable antipsychotics over oral ones in reducing hospitalizations.[16,17] The only of these to be published to date had rates of hospitalization (45% for oral medication and 39% for long-acting injectable over 2 years) that are similar to the current study.[16]

Nevertheless, the use of long-acting injectable antipsychotic medications is supported by some systematic reviews[18,19] and expert panels[20,21] for outpatients at increased risk of relapse. A limitation is that the study did not include subjective measures of medication satisfaction or global well-being. In addition, this study did not address current cost differences for payers, which may be substantial as paliperidone palmitate is still on patent while haloperidol decanoate is available as a generic drug.

Conclusions

Among adults with schizophrenia or schizoaffective disorder, treatment with paliperidone palmitate, compared with haloperidol decanoate, did not result in a statistically significant difference in efficacy failure, but the results do not rule out the possibility of a clinically meaningful difference. The results are consistent with previous research that has not found large differences in the effectiveness of newer and older antipsychotic medications.

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